



Figure 1: Vdbench

WD name must be unique.

WD parameters include the following:

- sd – Test device
- seekpct – Percentage time to move location
- rdpcct – Read percentage
- xfersize – Transfer size
- skew – Percentage of skew this workload receives from the total I/O rate
- wd – Default setup for the workload
- threads – How many concurrent operations for this workload
- hotband – Executes hot band workload against a range of storage

```
wd=hotwd_uniform,skew=6,sd=sd*,seekpct=100,rdpct=50 wd=hotwd_hot1,sd=sd*,skew=28,seekpct=rand,hotband=(10,18)
```

Run definition

RD defines what storage and workload will be run together and for how long. Each run definition name must be unique.

RD parameters include:

- wd – Workload load definition
- iorate – Either IOPS, 'max' or 'curve'
- warmup – Warm-up time that will be excluded from the elapsed time (10s/2m/1h)
- elapsed – How long to run
- interval – Stats collection interval
- threads – Number of threads
- forrdpct – Range of percentage read to execute

```
rd=rd1_hband,wd=HOTWd*,iorate=MAX,warmup=30,elapsed=6H,interval=10,
pause=30,th=200 rd=rd1_seq,wd=wd_seq,iorate=MAX,forrdpct=(0,100),
xfer=256K,warmup=30,e1=20m,in=5,th=20
```

Here's a sample configuration file.

```
// SD: Storage Definition
// WD: Workload Definition
// RD: Run Definition

sd=sd1,lun=/dev/sdb0
wd=rr,sd=sd1,xfersize=4096,rdpct=100
rd=run1,wd=rr,iorate=100,elapsed=10,interval=1

// Single raw disk, 100% random read of 4k records at i/o
rate * of 100 for 10 seconds
```

To run the Vdbench, type the following commands:

```
/vdbench/vdbench -f conf1.txt conf2.txt -o output+
/vdbench/vdbench -i 10 -f simple_test.conf -o simple_test+
```

Parameters include:

- -f – Configuration file(s)
- -o – Output directory
- -e – Elapsed time override
- -i – Interval time to override
- -w – Warm-up time to override
- -j – Activates data validation and journaling

Utility functions

- */vdbench sds* – This is used to generate SD params.
- */vdbench dvpost* – This is used to analyse data corruption issues.
- */vdbench rsh* – This is to start Vdbench slaves on the host and return *stdout* and *stderr*.
- */vdbench compare* – Compares two sets of Vdbench output directories.

Reports

On completing a successful test run, Vdbench creates the following HTML files.

Summary.html – Reports the total workload generated for each interval.

Totals.html – Reports only the total without detailed interval information.

Logfile.html – Reports Vdbench logging information for debugging.

Kstat.html – kstat statistics (only on Solaris).

Histogram.html – Reports distribution of response times for both reads and writes combined.

Flatfile.html – Performance data file, which can be used to generate performance charts. 

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